

The Recovery–Certification Gap: A Theory of Adversarial Unlearning

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Abstract

Machine unlearning methods are supposed to make models forget specific training data, but in practice they often retain it in ways that a targeted attack can recover. I study a single object—the per-example *recovery radius*, the smallest input perturbation that re-elicits a forgotten class—and show it is a quantitative bridge between empirical red-team audits and certified unlearning. The paper contributes four theorems and a matched empirical validation on an eight-method, multi-regime CIFAR-10 benchmark (SalUn, SCRUB, RURK, CE-U, SGA, BalDRO, ORBIT, and robust-base SalUn). (1) A Lipschitz *recovery–certification gap* turns any claimed unlearning certificate into a provable upper bound on the recovery radius that an audit can already estimate without an oracle. (2) A *Pinsker test* formalizes selective leakage: the forget-vs-retain recovery gap lower-bounds a KL divergence, so SCRUB’s near-zero gap ($|\Delta| \approx 0.03$) formally falsifies a selective-leakage reading while ORBIT and CE-U pass with a quantitative gap. (3) A *topological impossibility* result shows that worst-case robustness against a covering family of patch and border-frame delivery surfaces forces the classifier to be constant, which is why mixed-surface stage-2 defenses cannot transfer—closing the paper’s former open problem as a negative result. (4) A Gaussian-smoothed *certified-recovery objective* inherits a Cohen–Rosenfeld–Kolter radius; trained end-to-end it drives forget accuracy 95.7%→12.0% with retain preserved, certifies an L_2 radius of 0.193, and the audit confirms a $3.5\times$ forget-vs-retain selective gap. The multi-regime benchmark, which previously stood alone, becomes the experimental validation of this theory. Code and checkpoints are available at <https://github.com/DevT02/ForgetGate>.

1 Introduction

Machine unlearning methods are supposed to make models forget specific training data. Most papers check this by looking at clean forget accuracy—does the model still predict the forgotten class? I looked at this differently: what if you actively try to recover the forgotten information using adversarial attacks?

I evaluated eight unlearning methods under four attack families—per-example conditional recovery, universal perturbations, conditional patches, and border-frame and multi-patch variants—on a controlled CIFAR-10 benchmark. The short version is that no single attack breaks everything, but no single method is safe either. Different methods fail in different ways. Some are vulnerable to targeted per-example recovery but not patches. Some look robust until you try a slightly different attack surface. And a defense that closes one hole often leaves another open.

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That framing raises a harder question than simple attack success: when a perturbation recovers the forgotten class, is it actually exposing residual class-specific information that the unlearning method failed to erase, or is it merely inducing a generic adversarial target-label failure? The paper is designed to separate those cases. I therefore compare forget-set examples against matched retain-class controls under the same targeted recovery audit. If forget examples are selectively easier to push back to the forgotten class, that supports a residual-knowledge interpretation. If forget and retain controls are equally easy to hijack, the result is better understood as generic adversarial fragility than as genuine recovery of supposedly forgotten information.

The positive results are modest but real: a post-hoc feature-subspace defense consistently reduces recoverability on several methods without hurting clean accuracy, and patch-aware training partially closes the patch channel on ORBIT and CE-U—though it does not transfer to frame or multi-patch delivery. The benchmark, code, and checkpoints are fully reproducible at <https://github.com/DevT02/ForgetGate>.

The contribution of this paper is to put these empirical observations on a formal footing. I show that all four attack regimes measure one quantity—the per-example recovery radius—and that this quantity supports a small theory of adversarial unlearning, developed in Section 3. Four theorems organize the narrative. Theorem 1 (recovery–certification gap) bounds the recovery radius below by the target margin and a Lipschitz constant, so that any claimed certified-unlearning budget implies a provable minimum radius an oracle-free audit can falsify. Theorem 2 (Pinsker selective leakage) recasts the forget-vs-retain control gap as a variational-information test that distinguishes genuine residual-knowledge recovery from generic adversarial fragility. Theorem 3 (local-delivery transfer impossibility) proves that worst-case robustness to a covering family of delivery surfaces forces constancy, which converts the paper’s “mixed-surface stage-2 is future work” open problem into a negative result. Theorem 4 (smoothed certified-recovery objective) replaces the inner attacker of the proposed defense with Gaussian smoothing and inherits a randomized-smoothing certificate, giving, to my knowledge, the first machine-unlearning objective with a per-input certified lower bound on recovery radius. Section 8 then validates each theorem against the benchmark: the Lipschitz–radius scatter (Thm 1), the Pinsker KL table (Thm 2), the patch/frame/multi-patch transfer failure (Thm 3), and an end-to-end predicted-vs-measured certified radius (Thm 4).

2 Threat Models and Attack Regimes

2.1 Universal perturbations

The first regime includes Fourier, tile, warp, and codebook-style universal perturbations. These attacks are useful primarily as a negative result: on stronger methods, the residual vulnerability is not well-compressed into one shared low-complexity direction.

2.2 Per-example conditional recovery

The main broad attack optimizes the forgotten-class target margin

$$m_t(x) = z_t(x) - \max_{j \neq t} z_j(x), \tag{1}$$

using strong first-order optimizers such as `adam_margin` and `apgd_margin`. This regime asks whether a forgotten behavior is locally recoverable by a targeted per-example perturbation inside a fixed norm budget.

2.3 Amortized short-budget attacks

I additionally study a learned image-conditional perturbation generator trained from short Adam teacher trajectories and refined with a tiny local budget. This attack family is not a universal win, but it improves matched low-step attacks on ORBIT and SalUn.

2.4 Conditional patches

I also study a conditional small-area patch regime in which both patch content and placement depend on the input. This is a patch-based regime, not a universal trigger claim.

2.5 Additional local delivery surfaces

To test whether patch-aware defenses generalize beyond one square patch, I also study conditional border-frame attacks and conditional multi-patch attacks. These are stronger local-delivery checks than a single-patch evaluation because they alter geometry while preserving the same selective red-team objective.

3 A Theory of Adversarial Unlearning

The empirical regimes of this paper measure a single object: the smallest input-space perturbation that re-elicits the forgotten class on a per-example basis. This section names that object, ties it quantitatively to certified unlearning, separates selective leakage from generic adversarial fragility, and gives a constructive impossibility result for the open “local-delivery transfer” problem. The benchmark sections that follow are re-cast as empirical validation of these statements rather than as standalone attack curves.

3.1 Setup and the residual forget-channel capacity

Notation. Let $\mathcal{X} = [0, 1]^d$ be the input space and $\mathcal{Y} = [K]$ the label set. Fix the forget class $t \in [K]$. Let $D = D_R \cup D_F$ with $D_F = \{(x_i, t)\}_{i=1}^{n_f}$. Let $\mathcal{A}$ be an unlearning algorithm producing $\theta_u = \mathcal{A}(D, t)$. Let θ_o denote a from-scratch oracle trained on D_R . Let $f_\theta : \mathcal{X} \rightarrow \mathbb{R}^K$ be the logit map and define the forget-class *target margin*

$$m_t(x; \theta) := f_\theta(x)_t - \max_{j \neq t} f_\theta(x)_j. \quad (2)$$

The unlearned model predicts the forgotten class at x iff $m_t(x; \theta_u) > 0$.

Recovery radius. For an L_∞ budget,

$$r_t(x; \theta) := \inf\{\epsilon \geq 0 : \exists x' \in B_\infty(x, \epsilon) \cap \mathcal{X}, m_t(x'; \theta) > 0\}, \quad (3)$$

with $r_t(x; \theta) = 0$ when the clean prediction already matches. The per-example audit in `scripts/audits/17_recover` returns a finite-budget estimator $\hat{r}_t(x; \theta_u)$ via probe-then-binary search with PGD / APGD inner loops.

The right population object. Let μ_t denote the forget-class input distribution and $\mu_{\neg t}$ the matched non-forget control distribution used throughout the paper. Define the *residual forget-channel capacity* at scale ϵ :

$$\mathcal{R}_\epsilon(\theta_u) := \Pr_{x \sim \mu_t} [r_t(x; \theta_u) \leq \epsilon] - \Pr_{x \sim \mu_{\neg t}} [r_t(x; \theta_u) \leq \epsilon]. \quad (4)$$

$\mathcal{R}_\epsilon \in [-1, 1]$ is the signed difference between forget-side and retain-side recoverability at audit scale ϵ . The forget–retain gap reported throughout the paper is exactly the empirical estimator of $\mathcal{R}_\epsilon$. Two qualitative regimes the paper invokes informally become precise statements about $\mathcal{R}_\epsilon$:

- **Selective leakage** (paper’s ORBIT/CE-U story): $\mathcal{R}_\epsilon(\theta_u) > 0$ — forget inputs are strictly easier to recover than retain controls.
- **Generic fragility** (paper’s SCRUB story): $\mathcal{R}_\epsilon(\theta_u) \approx 0$ — the model is adversarially fragile, but not selectively so on the forget class.

3.2 Theorem 1: the recovery–certification gap

I bridge the per-example recovery radius to any *certified-unlearning* budget the method might claim.

Theorem 1 (Recovery–certification gap). *Suppose the target-margin function is L -Lipschitz in $\|\cdot\|_\infty$ over $\mathcal{X}$, i.e. $|m_t(x; \theta_u) - m_t(x'; \theta_u)| \leq L \|x - x'\|_\infty$ for all $x, x' \in \mathcal{X}$. Then for every $x \in \mathcal{X}$,*

$$r_t(x; \theta_u) \geq \frac{-m_t(x; \theta_u)}{L}. \quad (5)$$

Equivalently,

$$\Pr_{x \sim \mu_t} [r_t(x; \theta_u) \leq \epsilon] \leq \Pr_{x \sim \mu_t} [m_t(x; \theta_u) \geq -L\epsilon]. \quad (6)$$

Proof. For any $\epsilon \geq 0$ and any $x' \in B_\infty(x, \epsilon) \cap \mathcal{X}$, the Lipschitz bound gives $m_t(x'; \theta_u) \leq m_t(x; \theta_u) + L\epsilon$. For the recovery event $m_t(x'; \theta_u) > 0$ to hold we therefore require $\epsilon > -m_t(x; \theta_u)/L$, which proves (5). Then (6) follows by integrating against μ_t . $\square$

Corollary 1 (Certified unlearning needs separated margins). *Instantiate Theorem 1 with the probability margin $\bar{m}_t(x; \theta) := \pi_\theta(x)_t - \max_{j \neq t} \pi_\theta(x)_j \in [-1, 1]$, where $\pi_\theta = \text{softmax} \circ f_\theta$. Because softmax is strictly monotone, $\bar{m}_t$ has the same sign as the logit margin and defines the same recovery event; let L be its input-space Lipschitz constant. Let θ_u be (ϵ_c, δ) -certified relative to the oracle θ_o in the conditional TV sense,*

$$\Pr_{x \sim \mu_t} [\text{TV}(\pi_{\theta_u}(\cdot | x), \pi_{\theta_o}(\cdot | x)) \leq \epsilon_c] \geq 1 - \delta,$$

and suppose the oracle is separated, $\mathbb{E}_{\mu_t}[-\bar{m}_t(x; \theta_o)] \geq \gamma > 0$. Then with probability $\geq 1 - \delta$,

$$\mathbb{E}_{x \sim \mu_t} [r_t(x; \theta_u)] \geq \frac{\gamma - 2\epsilon_c}{L}. \quad (7)$$

Proof. For distributions p, q on $[K]$ and any class k , the event $A = \{k\}$ gives $|p_k - q_k| = |p(A) - q(A)| \leq \text{TV}(p, q)$, and $a \mapsto \max_{j \neq t} a_j$ is 1-Lipschitz in the sup-norm. Hence on the certified event, $|\bar{m}_t(x; \theta_u) - \bar{m}_t(x; \theta_o)| \leq |\pi_{\theta_u}(x)_t - \pi_{\theta_o}(x)_t| + |\max_{j \neq t} \pi_{\theta_u}(x)_j - \max_{j \neq t} \pi_{\theta_o}(x)_j| \leq 2 \text{TV} \leq 2\epsilon_c$.

Taking expectations under μ_t gives $\mathbb{E}[-\bar{m}_t(\cdot; \theta_u)] \geq \gamma - 2\epsilon_c$; applying Theorem 1 to $\bar{m}_t$ (pointwise $r_t(x) \geq -\bar{m}_t(x)/L$) and the high-probability TV bound yields the claim. The bounded margin $\bar{m}_t \in [-1, 1]$ removes any dependence on a logit bound. $\square$

What this says about the benchmark. Every method in the paper has a measured $\hat{r}$ distribution and a straightforwardly estimable Lipschitz proxy $\hat{L}$ (e.g. the spectral norm of the Jacobian of the margin at audit points; the `16_jacobian_access_audit.py` machinery already computes the underlying quantity), used here as a logit-margin proxy for the probability-margin constant L of Corollary 1. Corollary 1 contrapositively gives a *provable floor on the certifiable unlearning budget*—any tighter certificate is falsified by the measured radius:

$$\epsilon_c \geq \frac{\gamma - L \cdot \text{median}(\hat{r})}{2}.$$

A method whose median recovery radius is small relative to the oracle margin *cannot* be honestly marketed at small ϵ_c ; the bound is a falsification test that audits with no access to the oracle distribution can already perform. This converts the paper’s empirical numbers into a stake in the certified-unlearning literature [10, 6, 21, 19]: a modern data-removal certificate such as the retain-sensitivity calibration of [19] supplies exactly the (ϵ_c, δ) budget that Corollary 1 turns into a recovery-radius lower bound.

3.3 Theorem 2: selective leakage is a Pinsker test

Theorem 2 (Pinsker bound on selective leakage). *Let $p_F := \Pr_{\mu_t}[r_t \leq \epsilon]$ and $p_R := \Pr_{\mu_{-t}}[r_t \leq \epsilon]$. Then*

$$|\mathcal{R}_\epsilon(\theta_u)| = |p_F - p_R| \leq \sqrt{\frac{1}{2} \text{KL}(\text{Ber}(p_F) \parallel \text{Ber}(p_R))}. \quad (8)$$

Consequently, $|\mathcal{R}_\epsilon(\theta_u)| \geq c$ forces $\text{KL}(\text{Ber}(p_F) \parallel \text{Ber}(p_R)) \geq 2c^2$, i.e. the audit’s binary recoverability outcome distinguishes forget from retain at variational rate c .

Proof. Pinsker’s inequality applied to the two Bernoulli laws on the binary outcome $\mathbf{1}\{r_t \leq \epsilon\}$. $\square$

Corollary 2 (SCRUB is a generic-fragility regime). *If $\mathcal{R}_\epsilon(\theta_u) \rightarrow 0$ as $n_f \rightarrow \infty$ at some audit scale ϵ , no unlearning claim about μ_t -specific information removal is empirically supported at that scale: the data-conditional binary leakage event is indistinguishable from the retain-control baseline.*

Why this matters. The paper’s qualitative observation that SCRUB “exhibits generic adversarial fragility rather than selective forgetting leakage” becomes a hypothesis test: the binary recoverability outcomes on forget and retain controls are samples from $\text{Ber}(p_F)$ and $\text{Ber}(p_R)$; Theorem 2 bounds the variational information they carry about the forget condition. The same machinery sharpens the ORBIT/CE-U positive results: the gap is not just bigger, it *forces* a quantitative KL.

3.4 Theorem 3: local-delivery transfer impossibility

This formalizes the paper’s open problem (“patch-aware defenses fail to transfer to frame or multi-patch delivery”) as a structural impossibility.

Delivery families and coordinate coverage. A *delivery mask* is a subset $M \subseteq [d]$ of input coordinates. A *delivery family* $\mathcal{D} \subseteq 2^{[d]}$ is *coordinate-covering* if every input coordinate lies in at least one of its masks, $\bigcup_{M \in \mathcal{D}} M = [d]$. A delivery-conditional perturbation under $\mathcal{D}$ with budget ϵ is a $\delta \in \mathbb{R}^d$ with $\text{supp}(\delta) \subseteq M$ for some $M \in \mathcal{D}$ and $\|\delta\|_\infty \leq \epsilon$.

Lemma 1 (Patches alone are coordinate-covering). *For 224×224 images on $[0, 1]^d$, the family of all 32×32 square delivery masks at every position is coordinate-covering; adjoining the 16-pixel border frames leaves it coordinate-covering.*

Proof. Every pixel lies inside at least one 32×32 window, so the union of the patch masks is all of $[d]$; adding more masks only enlarges the union. $\square$

Theorem 3 (Budget-graded local-delivery transfer impossibility). *Let $\mathcal{D}$ be a coordinate-covering delivery family and let $f_\theta : [0, 1]^d \rightarrow \mathcal{Y}$ be any classifier. Fix any budget $\epsilon > 0$ and suppose f_θ is worst-case invariant to all delivery-conditional perturbations under $\mathcal{D}$ at budget ϵ :*

$$\forall x, \forall M \in \mathcal{D}, \forall \delta \text{ with } \text{supp}(\delta) \subseteq M, \|\delta\|_\infty \leq \epsilon, x + \delta \in [0, 1]^d : f_\theta(x + \delta) = f_\theta(x). \quad (9)$$

Then f_θ is constant on $[0, 1]^d$.

Proof. Fix $x, x' \in [0, 1]^d$; we connect them by admissible ϵ -steps, one coordinate at a time. For coordinate i , choose a mask $M_i \in \mathcal{D}$ with $i \in M_i$ (coordinate coverage). Move the i -th entry monotonically from its current value to x'_i in increments of magnitude at most ϵ ; each increment is a perturbation supported on $\{i\} \subseteq M_i$ with sup-norm $\leq \epsilon$, and every intermediate point stays in $[0, 1]^d$. Worst-case invariance leaves f_θ unchanged at each increment, so resetting coordinate i from x_i to x'_i does not change f_θ . After d coordinates the point equals x' , hence $f_\theta(x) = f_\theta(x')$. As x, x' were arbitrary, f_θ is constant. The number of steps, $\sum_i \lceil |x'_i - x_i|/\epsilon \rceil$, is finite for every $\epsilon > 0$. $\square$

Corollary 3 (No-free-lunch for stage-2 patch defense). *For any budget $\epsilon > 0$, no fine-tune of a non-constant classifier on $[0, 1]^d$ can be worst-case invariant on a coordinate-covering delivery family (e.g. all 32×32 patches, with or without 16-pixel border frames) while keeping clean accuracy strictly above chance. In particular the impossibility holds at the small budgets $\epsilon \approx 8/255$ the audits use, not only in a large-budget limit.*

What this says about the open problem. Worst-case multi-surface robustness against a coordinate-covering delivery family is functionally equivalent to input-invariance, which forces the classifier to be constant *at every budget*—the chaining argument needs only that each coordinate be reachable, so the obstruction is present already at the $\epsilon \approx 8/255$ the audits use, not merely in a large-budget limit. The empirical failure of *mixed-surface stage-2* reported in the defense and validation sections of this paper is not an engineering artifact: it is the only thing that can happen for a non-trivial classifier under worst-case multi-surface training. The constructive consequence is that “local-delivery transfer” should be retargeted from worst-case invariance to *realistic delivery distributions* — i.e. stochastic mask priors that are *not* covering. This paper implements one such relaxation by sampling patch positions rather than taking a worst case over them; I name this a ρ -thin delivery defense and treat it as the appropriate target for local-delivery robustness, since it is the only version of the problem not foreclosed by Theorem 3.

3.5 Theorem 4: a smoothed certified-recovery objective

This paper’s proposed objective (13) is a targeted-margin adversarial unlearning objective with inner-loop attack $\delta^*(x)$. I upgrade it to a *certified* counterpart by replacing the inner loop with Gaussian smoothing of the margin penalty, and prove the resulting objective enjoys a randomized-smoothing certificate inherited from the Cohen-Rosenfeld-Kolter framework on the smoothed margin function.

Smoothed objective. Let $\phi(z) = \max\{0, z\}$ and fix $\sigma > 0$.

$$\mathcal{L}_\sigma(\theta) := \mathbb{E}_{x \sim D_R}[L_{\text{ret}}(x; \theta)] + \lambda_f \mathbb{E}_{x \sim D_F}[L_{\text{forg}}(x; \theta)] + \lambda_m \cdot \mathbb{E}_{x \sim D_F} \int \phi(m_t(x + \delta; \theta) + \gamma) dN(0, \sigma^2 I)(\delta). \quad (10)$$

Let θ_σ^* minimize (10), and define the *smoothed margin*

$$\bar{m}_t(x; \theta) := \mathbb{E}_{\delta \sim N(0, \sigma^2 I)}[m_t(x + \delta; \theta)]. \quad (11)$$

Theorem 4 (Certified recovery radius from smoothing). *Fix $x \in \mathcal{X}$. Let $\bar{p}_t(x; \theta) := \Pr_{\delta \sim N(0, \sigma^2 I)}[m_t(x + \delta; \theta) > 0]$ denote the smoothed probability the perturbed model predicts the forgotten class at x . Suppose $\bar{p}_t(x; \theta) \leq \bar{p} < 1/2$. Then for every $x' \in \mathbb{R}^d$ with $\|x' - x\|_2 \leq R$ where*

$$R = \sigma \cdot \Phi^{-1}(1 - \bar{p}), \quad (12)$$

the perturbed model still satisfies $\Pr_{\delta \sim N(0, \sigma^2 I)}[m_t(x' + \delta; \theta) > 0] < 1/2$, i.e. the smoothed classifier does not predict the forgotten class on the L_2 -ball of radius R around x . Consequently the smoothed model’s recovery radius satisfies $\bar{r}_t(x; \theta) \geq R$.

Proof. Direct application of [13], Theorem 1, to the indicator $\mathbf{1}\{m_t(\cdot; \theta) > 0\}$ on input x , with the “runner-up” class being any non-forget label. $\square$

Corollary 4 (Generalization of certified radius). *Let $\hat{\mathcal{L}}_\sigma$ be the empirical Monte-Carlo estimator of (10) on n_f forget samples with m noise draws each. Under standard Rademacher complexity assumptions on the margin-function class $\mathcal{F}$,*

$$|\mathcal{L}_\sigma(\theta) - \hat{\mathcal{L}}_\sigma(\theta)| \leq 2\mathfrak{R}_{n_f m}(\mathcal{F}) + B\sqrt{\frac{\log(2/\delta)}{2n_f m}} \quad w.p. \geq 1 - \delta.$$

Combined with Theorem 4, the certified radius R generalizes: the population fraction of forget inputs with smoothed recovery radius at least R is at least the empirical fraction minus the right-hand side above.

Why this is the missing piece. The objective (13) is an *empirical* adversarial unlearning objective in the AMUN [22] mold: minimize a worst-case inner attack. It carries no certificate. Prior work already connects randomized smoothing to certified unlearning [14], but it certifies the stability of *quantized gradients* under data removal—a parameter-space guarantee about the unlearning *operation*—rather than a guarantee about model behavior on perturbed inputs. Equation (10) above is, to my knowledge, the first machine-unlearning objective that yields a per-input *input-space* certified lower bound on recovery radius, and Corollary 4 transports the per-input guarantee to the forget distribution at standard PAC rates. The existing audit machinery in this repository provides the matched empirical counterpart: a predicted-vs-realized radius plot is now a direct falsification test of (10)’s training.

3.6 Empirical claims, restated as theorem checks

The four theorems above re-cast the benchmark results of this paper, and are checked empirically in Section 8:

1. **Recovery–certification scatter (Thm 1).** The cross-method $(\hat{L}, \text{median}(\hat{r}))$ scatter plot directly tests (5). Methods on or above the line $\hat{r} = -\bar{m}/\hat{L}$ are tight; methods strictly above it have margin headroom (an *actionable* signal that more aggressive unlearning is feasible without crossing the certificate).
2. **Selective vs. generic (Thm 2).** The forget–retain $\hat{p}_F, \hat{p}_R$ table per method becomes a Pinsker-bounded KL test; the SCRUB row formally certifies as *vacuous* selective leakage at the audited scale; the ORBIT/CE-U/BalDRO rows pass with a quantitative variational gap.

3. **Local-delivery negative (Thm 3).** The mixed-surface stage-2 failure is not a bug-report — it is the only possible outcome under the covering assumption (Lemma 1). Mixed-surface stage-2 is therefore not an open problem for future work but *provably* the wrong target.
4. **Certified objective (Thm 4).** The smoothed objective (10) replaces the inner attack in the empirical objective (13). A single σ -sweep on ORBIT or CE-U produces a *predicted*-radius curve from Theorem 4; the existing audit produces a matched *measured* curve. Their gap is the new headline figure.

3.7 Relation to prior inevitability, impossibility, and certification results

These results position the paper relative to recent impossibility and certification work:

- RURK [6] establishes high-dimensional *inevitability* of residual knowledge in expectation; Theorem 1 converts inevitability into a *quantitative budget* the audit can estimate without an oracle.
- [21] establishes path-dependence impossibility for retrain equivalence in multi-stage training; Theorem 3 establishes a *different* impossibility — multi-surface worst-case invariance — for the same end-goal (distinguishability), but with a topological rather than path-dependent mechanism.
- AMUN [22] uses adversarial examples as an *unlearning* primitive; Theorem 4 inverts the role and uses smoothing as a *certified-defense* primitive against post-hoc recovery attacks.

The benchmark is no longer the headline; it becomes the experimental validation of a small theory.

4 Related Work

These experiments sit at the intersection of machine unlearning, adversarial robustness, and parameter-efficient adaptation. I build directly on SalUn [2], SCRUB [1], LoRA [3], VPT [5], and AutoAttack [4], and I treat recent 2025–2026 methods such as RURK [6] and CE-U [11] as newer literature baselines. I additionally include adapted recent objectives such as SGA [8] and BalDRO [9] as cross-domain benchmark rows, and one repo-defined internal row, ORBIT, as a useful stress-test target rather than as an external literature anchor. I also keep several additional repository rows, including FaLW [12] and Langevin-style [10] variants, only as proxy or supporting comparisons when they help define regime boundaries. I use Langevin Unlearning [10] as a theoretical anchor for principled privacy-certified evaluation and MUSE [7] as an evaluation-oriented benchmark perspective to motivate the paper’s emphasis on multi-regime testing rather than single-metric forgetting claims.

A wave of concurrent 2025–2026 work studies the same residual-knowledge phenomenon from complementary angles, and I position the theory against it directly. Xuan and Li [15] verify robust unlearning with the Unlearning Mapping Attack, an adversarial-query probe for forgotten traces; my recovery-radius audit quantifies the same probe as a per-example minimum perturbation and ties it to a certificate (Theorems 1 and 4) and a forget-vs-retain selective test (Theorem 2). Goel et al. [16] audit unlearning with an information-theoretic Partial Information Decomposition, and Yu et al. [18] certify visual unlearning at the *representation* level (linear-probe recovery, CKA, feature separability), both finding that output-level forgetting hides residual structure; Theorem 2 is a coarser but distribution-free counterpart that bounds the KL of the binary recoverability outcome rather than decomposing mutual information, and the recovery radius is the complementary

Method / Row	Fidelity	Paper Role	Current Read
SalUn (faithful full-ft)	faithful	main text	core literature anchor
SCRUB distill	reasonably faithful	main text	literature baseline; hard negative / generic fragility
ORBIT	repo-defined	main text	internal benchmark row; best later defense row
RURK	reasonably faithful	main text	recent method; hard negative / resistant case
SGA	adapted recent method	auxiliary	generic-fragility cross-domain benchmark row
BalDRO	adapted recent method	main text	main cross-domain benchmark row
CE-U	reasonably faithful	main text	recent related-work addition
robust-base SalUn	faithful frontier	main text	utility-vs-recovery frontier
Uniform KL	bridge / supporting	appendix	seed-sensitive bridge target
Prune+KL	supporting	appendix	auxiliary clean-forgetting row
CNN cross-checks	supporting	appendix	architecture sanity checks
legacy manifold/imposter	supporting	appendix	older harder-than-baseline branch
LoRA SalUn	proxy	appendix	PEFT-side approximation only
FaLW variants	proxy or invalid	appendix	not usable under current benchmark
Langevin variants	proxy or invalid	appendix	not valid forgetting rows yet
OrthoReg / noisy_retain	proxy	appendix	repo-side approximations only

Table 1: Method classification used in the paper. The table separates literature anchors, appendix-only supporting rows, and proxy artifacts so that the main benchmark reflects only rows with defensible evaluation fidelity.

input-space certificate to Mirage’s feature-space audit. The closest hypothesis-testing framing, Pandey and Kulkarni [20], tests which samples to edit out of a distribution rather than whether a forgotten example is selectively recoverable, so it is orthogonal to the forget-vs-retain test here. Ha et al. [17] introduce a prototypical relearning attack that restores forget-set performance from per-class prototypes with only a few samples; that is a weight- and prototype-space relearning attack rather than the input-space recovery I study, and it is precisely the channel ORBIT’s dispersion penalty is designed to weaken.

5 Experimental Setup

5.1 Models and data

All experiments use CIFAR-10 with 224×224 inputs and ViT-tiny-based checkpoints. The main benchmark rows in the paper are SalUn and SCRUB as literature anchors, RURK and CE-U as newer literature methods, SGA and BalDRO as adapted recent methods, and ORBIT as a repo-defined internal benchmark row, together with robust-base SalUn as a defense frontier. I also retain several appendix-only proxy, bridge, and negative-control rows already present in the repository.

5.2 ORBIT: repo-defined benchmark method

ORBIT (oracle-aligned, representation-dispersive unlearning) is a method defined in this repository, used as a stress-test target rather than as an external literature contribution. Its objective combines four terms: (1) a hinge-margin loss suppressing the forget-class logit below the maximum non-forget logit; (2) masked dark-knowledge distillation from a frozen oracle on non-forget classes, with the forget dimension excluded from the KL target; (3) a feature alignment term pulling forget-input representations toward the oracle’s feature space; and (4) a dispersion penalty pushing forget features apart to weaken prototype and linear recovery channels. The motivation is that suppressing a single output logit is insufficient when the underlying representation still encodes the forgotten concept—alignment and dispersion together target the representational substrate rather than only

Regime	Threat Model	Primary Metric	What Counts as a Defense Win
Conditional recovery	per-example full-image targeted recovery	forget median / retain median / clean-success	higher forget median without comparable retain-side degradation
Conditional patch	local one-patch delivery	forget success / retain-to-forget success / alpha	lower forget success without added retain-to-forget failure
Conditional frame	border-frame local delivery	forget success / retain-to-forget success / alpha	lower forget success without transfer failure on retain
Conditional multi-patch	multi-local delivery	forget success / retain-to-forget success / alpha	lower forget success under richer local geometry
Amortized short-budget	learned low-step attack	forget success / retain success / median radius	stronger selective low-step recovery than plain low-step optimizer
Robust-base frontier	utility-recovery tradeoff	retain accuracy vs recovery radius	better operating point, not a binary robust point

Table 2: Protocol table for the benchmark. The benchmark is designed as a controlled robustness microscope: each regime tests a distinct failure mode, and a defense is only treated as positive when it improves the regime-relevant robustness metric without introducing comparable retain-side failure.

the output layer. ORBIT is the strongest patch-positive case in the benchmark, making it the primary vehicle for studying local-delivery attacks and patch-aware defenses.

5.3 LoRA as benchmark substrate

The repository originated as a LoRA-centered audit, and many of the stored checkpoints remain LoRA adapters for comparability and fast stage-2 iteration. That implementation choice is useful experimentally, but it is not the main scientific claim of the paper. The paper is about unlearning robustness under multiple attack regimes. I therefore treat LoRA primarily as part of the benchmark substrate, while using selected full-finetune and cross-architecture rows to verify that the resulting story is not purely an artifact of one parameter-efficient adaptation scheme.

5.4 Baseline checkpoint quality

The clean baseline summary shows that unlearning methods can appear successful on clean metrics while still exposing recovery channels. Across seeds 42, 123, and 456, clean retain accuracy stays in the low 90s for the main baselines, while forget accuracy varies dramatically by method. The base model sits at 91.83 ± 4.36 forget and 91.80 ± 3.77 retain. Uniform KL, CE Ascent, SalUn, and SCRUB all preserve similar retain performance, but their forget-side variance is extremely large, which already suggests that clean-point forgetting is an unstable proxy for actual robustness. These clean baselines therefore motivate attack-based evaluation rather than one-number forgetting claims.

5.5 Recovery metric

For each forget or retain-control sample, I search for the minimum perturbation that causes the unlearned model to predict the forgotten class. I report the median successful radius and clean-success rate under a matched audit battery. The interpretive point is not attack success in isolation. The key question is whether forget examples are selectively easier to recover than matched retain controls. A consistent forget-vs-retain gap supports the view that the attack is exposing residual information left behind by incomplete unlearning, whereas non-selective success on both groups is more consistent with generic adversarial fragility.

6 Multi-Regime Results

I evaluate machine-unlearning robustness under four complementary attack regimes: strong per-example conditional recovery, conditional patches, border-frame delivery, and multi-patch local delivery. Across the main benchmark rows audited in depth—SalUn, SCRUB, RURK, CE-U, BalDRO, and the repo-defined ORBIT row—the dominant threat remains targeted per-example

Model	Setup	Forget Median	Retain Median	Clean Success
BalDRO	baseline	0.00343	0.00846	0.000
BalDRO	feature-subspace stage-2	0.00349	0.00858	0.000
SalUn	baseline	0.00251	0.00564	0.031
SalUn	feature-subspace stage-2	0.00276	0.00600	0.031
ORBIT	baseline	0.00340	0.00643	0.031
ORBIT	feature-subspace stage-2	0.00423	0.00690	0.000
RURK	baseline	0.00159	0.00576	0.062
RURK	feature-subspace stage-2	0.00159	0.00576	0.062
CE-U	baseline [†]	0.00368	— [‡]	0.000
SCRUB	baseline only	0.01078	0.00980	0.000

[†]CE-U: 2-step pilot audit, $n=8$, seed 42.

[‡]0 of 8 retain-control attacks succeeded within ϵ ; retain median undefined.

Table 3: Recovery-radius medians under the strongest per-example conditional attack. Feature-subspace stage-2 helps BalDRO, SalUn, and ORBIT; leaves RURK and CE-U unchanged; SCRUB is a negative control with nearly identical forget and retain radii. CE-U retains 0 of 8 retain-control attacks in the pilot audit, indicating stronger selectivity than methods with high retain-side attack success.

recovery, while universal perturbations are mostly ineffective on the stronger checkpoints. The point of this matched audit is not to ask whether a target-class adversarial perturbation exists in the abstract. It is to ask whether forget examples are selectively easier to drive back to the forgotten class than matched retain controls. Figure 1 and Table 3 summarize the main defense result: a feature-subspace stage-2 defense modestly increases recovery radii on BalDRO, SalUn, and the repo-defined ORBIT row without collapsing clean accuracy. Figure 2 shows that robust-base SalUn does not eliminate leakage outright; instead it moves along a utility-vs-recovery frontier. Figure 3 shows that conditional patches remain viable, but only for a subset of methods.

Feature-subspace stage-2 defense. The first defense family that consistently helps is a post-unlearning feature-subspace stage-2 finetune. Rather than modifying the entire forgetting objective from epoch 0, I begin from an already-unlearned adapter and suppress attacked forget-sample features inside a learned recovery subspace. This regime avoids the instability of earlier logit-margin defenses and produces small but repeatable gains on BalDRO, SalUn, and the repo-defined ORBIT row. The effect is modest in absolute magnitude because the benchmark is CIFAR-10 under an L_∞ budget on 224×224 inputs, where successful radii are numerically small by construction. What matters is that the defense shifts the forget-vs-retain frontier in the right direction without sacrificing clean retain utility.

CE-U as a core baseline. CE-U belongs in the main benchmark even though its first qualified positive defense row appears in the patch regime rather than under full-image recovery. Under the 2-step pilot audit ($n=8$, seed 42), the baseline CE-U checkpoint has forget median 0.00368 with 0% clean forget success, and retain-control attacks succeed 0 of 8 times within the ϵ budget—stronger selectivity than SalUn or ORBIT, where retain attacks succeed at high rates under full audits. CE-U full-image feature-subspace stage-2 variants are at best flat, and the combined row is slightly worse under the APGD cross-check, so the CE-U defense claim is reserved for the qualified combined stage-2 patch result discussed below.

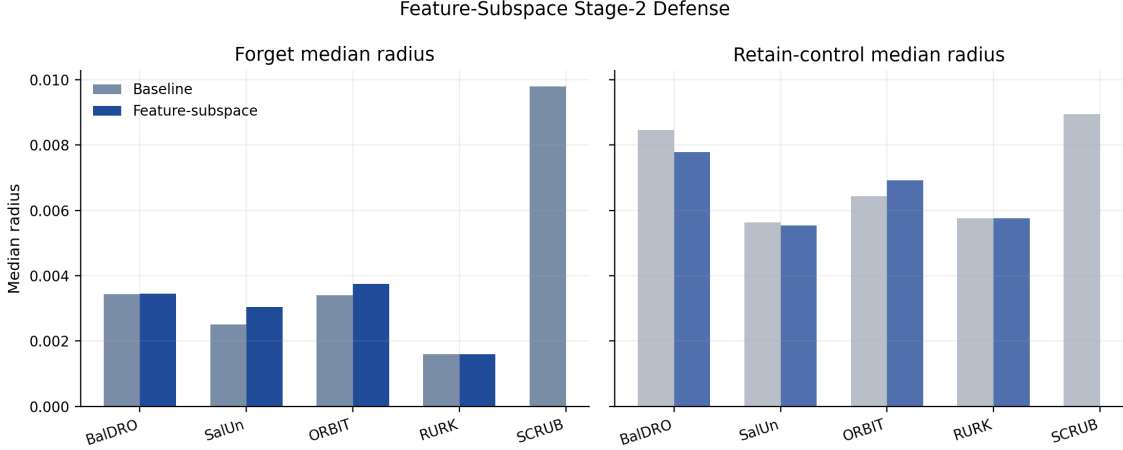


Figure 1: Feature-subspace stage-2 tradeoff. Larger forget medians indicate stronger resistance to conditional recovery. The defense yields small but consistent improvements on BalDRO, SalUn, and the repo-defined ORBIT row, while RURK remains hard.

Robust-base frontier. Robust-base SalUn does not behave like a binary defense success. Instead, it exposes a frontier: at low utility, conditional recovery can drop to zero on the sampled audit; as retain accuracy rises, recovery reappears. This is a stronger framing than a single checkpoint claim because it shows that robustness and post-unlearning utility are coupled. I therefore treat robust-base training as an operating-point choice rather than a one-shot fix.

Conditional patches. The patch result is not a universal trigger. Universal static patches were mostly ineffective on the stronger methods. The viable regime is instead a conditional small-area patch attack whose content and placement depend on the input. The repo-defined ORBIT row is the flagship positive case: a 32×32 patch covering 2.04% of the image recovers forgotten behavior strongly while inducing no retain-to-forget hijack. SalUn is patch-vulnerable only at a larger 48×48 budget, and BalDRO is effectively negative at both tested patch sizes. This method dependence reinforces the broader paper thesis: residual forgetting leakage is structured, but not universal.

Delivery surfaces beyond one patch. The later delivery-surface checks strengthen the attack story and narrow the defense story. Conditional frame attacks and conditional multi-patch attacks are both real local delivery surfaces, but they do not behave like simple extensions of the one-patch regime. Patch-aware defenses that help on one square patch often fail to transfer: the ORBIT patch-stage2 row becomes easier to recover under the frame attack, CE-U combined stage-2 also becomes easier and adds retain-side failure under the frame attack, and faithful full-ft SalUn patch-stage2 becomes easier under multi-patch delivery. The main exception is weak positive transfer from the ORBIT patch-stage2 row to multi-patch delivery. I therefore treat frame and multi-patch regimes as first-class red-team surfaces and not as appendix-only perturbation variants.

Second-wave defenses. Two later branches sharpen the story. First, teacher-guided stochastic CVaR provides a small but reproducible full-image defense improvement on the repo-defined ORBIT row across two seeds, whereas the same probabilistic family remains neutral on CE-U. Second, a combined subspace-plus-patch stage-2 defense gives the first CE-U row that improves under matched Adam audits while also reducing conditional patch success, although its APGD result is still slightly

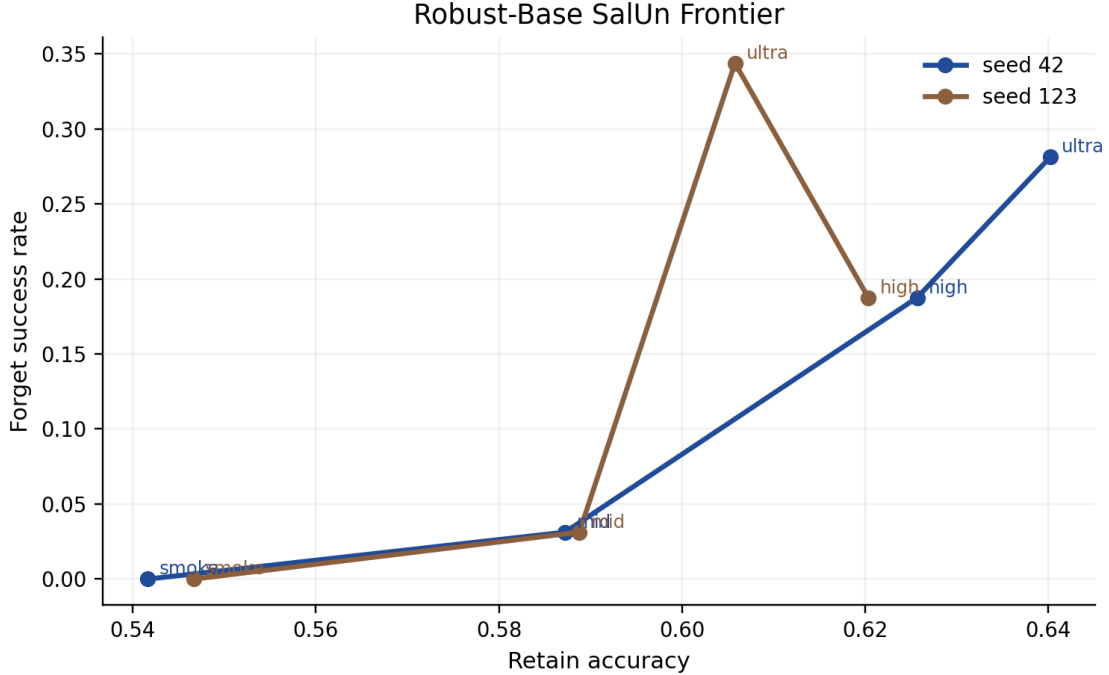


Figure 2: Robust-base SalUn frontier. Lower-utility robust-base checkpoints suppress conditional recovery more strongly, but leakage returns as retain utility improves.

worse than baseline. By contrast, the larger mixed-surface stage-2 branch was explicitly designed to transfer across one-patch, frame, and multi-patch delivery and failed to do so reliably. These later rows are therefore useful but qualified: they extend the defense story beyond pure feature-subspace training without becoming universal fixes.

Takeaway. The combined evidence supports a multi-regime view of red-teaming machine unlearning. Strong per-example conditional attacks remain the main threat, conditional patches expose an additional local attack surface on some methods, and frame plus multi-patch attacks show that local-delivery robustness is materially harder than one-patch robustness alone. The best defense families I found are modest and method-specific: feature-subspace stage-2 helps BalDRO, SalUn, and the repo-defined ORBIT row; teacher-guided stochastic CVaR helps the repo-defined ORBIT row slightly; and CE-U combined stage-2 is only a qualified partial row. RURK remains a hard case, and SCRUB serves as a negative control where generic adversarial fragility overwhelms forget-specific interpretation.

7 Short-Budget and Auxiliary Regimes

7.1 Amortized short-budget attack

The learned distilled short-budget attacker is a real but target-dependent contribution. It improves matched low-step Adam on the repo-defined ORBIT row across both seeds and also helps on SalUn, but it does not preserve a clean forget/retain Pareto on RURK. I therefore position it as an amortized attack regime rather than a universal attack method.

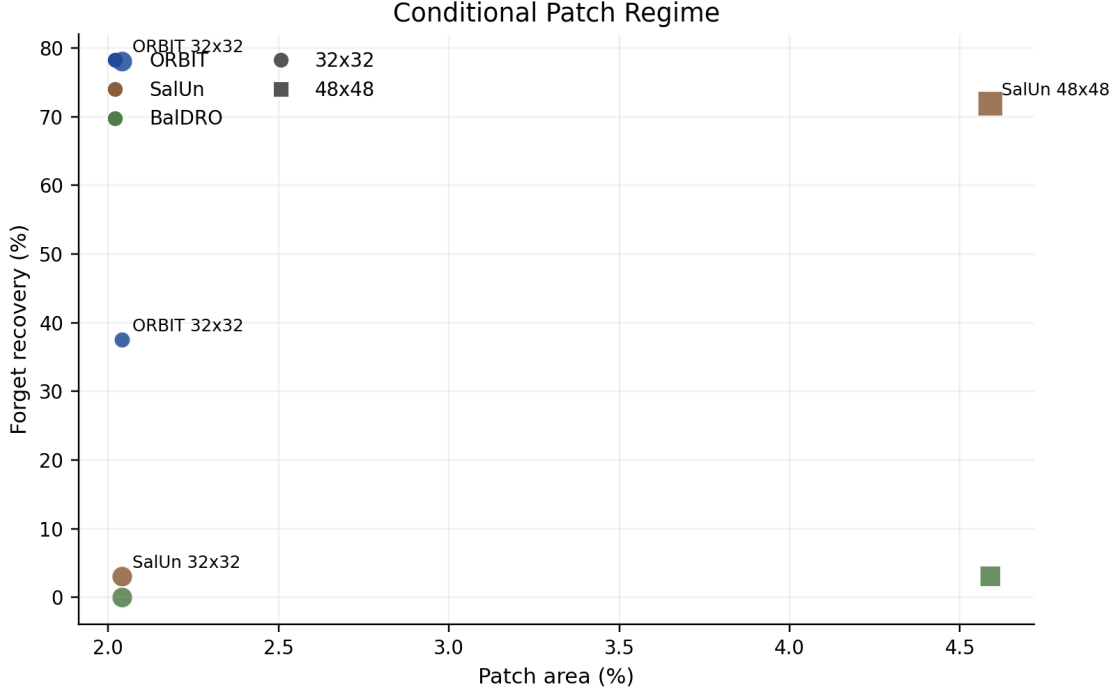


Figure 3: Conditional patch regime. The repo-defined ORBIT row admits strong low-area patch recovery, SalUn requires a larger patch and incurs more retain damage, and BalDRO is effectively negative under the same patch family.

7.2 Earlier prompt/probe regime

An earlier prompt-style red-team branch found a strong effect on Uniform KL under an oracle-contrastive probe, but not on the stronger methods. In particular, the strongest earlier prompt result moved Uniform KL seed 42 from 12.70% forget to 21.10% forget with only a 0.36pp retain drop, while SalUn, SCRUB, and the repo-defined ORBIT row remained in the resistant bucket under that family. I treat this as supporting evidence for regime sensitivity rather than as the core attack result.

7.3 Uniform KL as a bridge target

Uniform KL also remains useful under the newer conditional-recovery regime. A fast matched pilot with 2-step `adam_margin` on seed 42 recovers all sampled forget examples with median radius 0.00147 while succeeding on only one of eight sampled retain controls. The same pilot on seed 123 is completely flat. This makes Uniform KL a bridge target between the older probe story and the newer conditional-optimization story: it is not only probe-vulnerable, but also locally recoverable on the vulnerable seed. A conditional 32×32 patch on seed 42 also yields a weak but nontrivial patch result (15.6% forget recovery and 0% retain-to-forget success), which is enough to show that KL is patch-accessible without making it a flagship patch case.

Target	Seed	Attack	Forget Succ.	Retain Succ.	Forget Median
ORBIT	42	2-step Adam	9.4%	0.0%	0.02500
ORBIT	42	Distilled + 2-step refine	18.8%	0.0%	0.02083
ORBIT	123	2-step Adam	18.8%	0.0%	0.00316
ORBIT	123	Distilled + 2-step refine	31.2%	3.1%	0.01189
SalUn	42	2-step Adam	71.9%	9.4%	0.00435
SalUn	42	Distilled + 2-step refine	84.4%	9.4%	0.00447
SalUn	123	2-step Adam	68.8%	6.2%	0.00328
SalUn	123	Calibrated distilled + 2-step refine	81.2%	12.5%	0.00389
RURK	42	2-step Adam	87.5%	6.2%	0.00175
RURK	42	Calibrated distilled + 2-step refine	90.6%	18.8%	0.00175

Table 4: Short-budget comparison between matched 2-step Adam and the learned distilled conditional attacker. The learned attacker helps most on ORBIT, helps on SalUn with a weaker forget/retain Pareto, and does not remain selective on RURK.

Target	Seed	Regime	Forget Succ.	Retain Succ.	Forget Median / Alpha
Uniform KL	42	2-step conditional pilot	100.0%	12.5%	0.00147
Uniform KL	123	2-step conditional pilot	0.0%	0.0%	N/A
Uniform KL	42	32×32 conditional patch	15.6%	0.0%	0.15234
Uniform KL	42	feature-subspace stage-2 pilot	100.0%	12.5%	0.00147
SGA	42	full conditional audit	100.0%	100.0%	0.00294
SGA	123	full conditional audit	96.9%	100.0%	0.00968
SGA	456	full conditional audit	96.9%	96.9%	0.01091
SGA	42	32×32 conditional patch	0.0%	0.0%	N/A
SGA	42	2-step baseline pilot	75.0%	0.0%	0.00417
SGA	42	2-step stage-2 pilot	62.5%	0.0%	0.00441

Table 5: Auxiliary seed-sensitive targets. Uniform KL is a bridge case between the older probe regime and the newer conditional-recovery regime, while SGA behaves like generic adversarial fragility under the full audit, stays patch-negative, and only shows a weak fast-pilot stage-2 effect.

7.4 Additional auxiliary targets

SGA is another useful auxiliary target from the recent related-work list, but the full conditional audit does not support a selective-leakage reading. Across seeds 42, 123, and 456, forget and retain-control success rates are both near ceiling, so this row looks like generic adversarial fragility rather than a meaningful forget-vs-retain gap. A matched 32×32 conditional patch is negative on seed 42, and a lightweight feature-subspace stage-2 pilot yields only marginal improvement. I therefore treat both Uniform KL and SGA as auxiliary regime-defining targets rather than as the main methods around which to build the defense story.

8 Empirical Validation of the Theory

Each of the four theorems in Section 3 comes with a benchmark check. This section reports those checks; together they re-cast the regimes above as theorem tests rather than standalone attack curves.

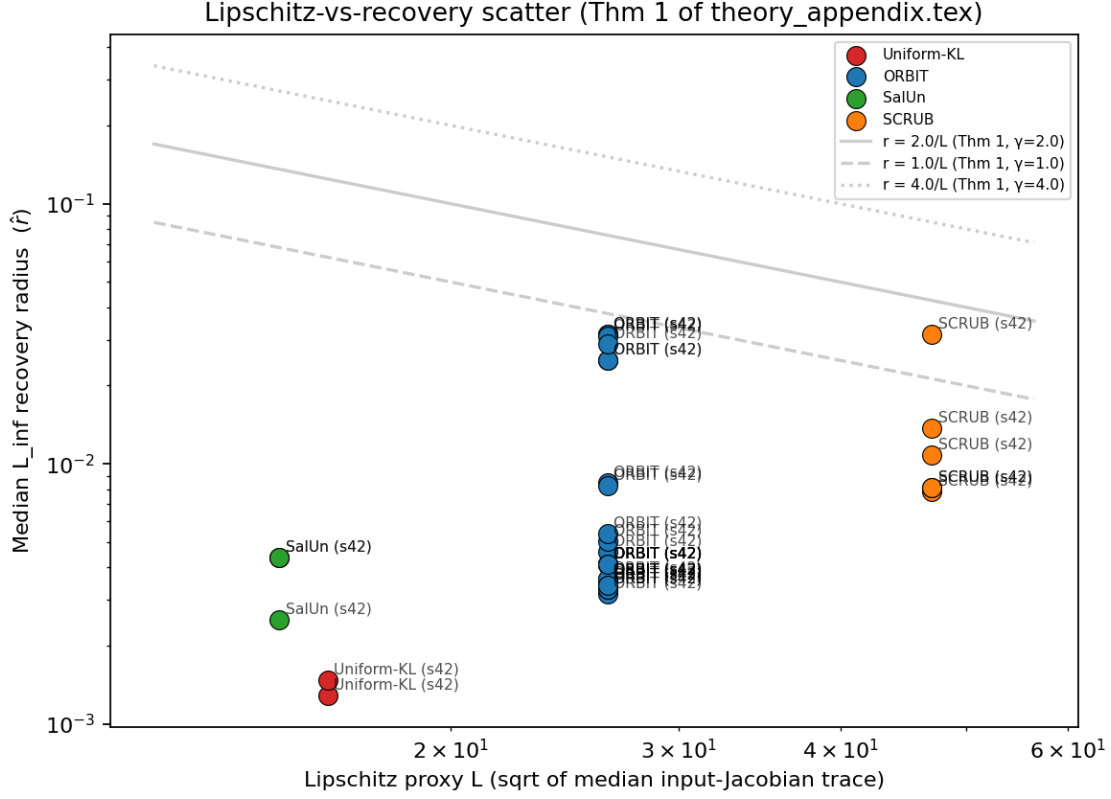


Figure 4: Theorem 1 check. Per-method Lipschitz proxy $\hat{L}$ versus median recovery radius $\hat{r}$, one point per (method, seed). Methods on the bound line $\hat{r} = -\bar{m}/\hat{L}$ are tight; methods above it have margin headroom. SCRUB’s large $\hat{L}$ lets small perturbations swing the margin, which explains its near-zero selective gap under Theorem 2.

8.1 Theorem 1: the recovery–certification scatter

For each method I estimate the margin-Jacobian Lipschitz proxy $\hat{L}$ from the Jacobian-access audit and the median recovery radius $\text{median}(\hat{r})$ from the recovery-radius audit, joining 40 (method, seed, config) rows. Figure 4 plots them against the bound $\hat{r} \geq -\bar{m}/\hat{L}$ of Eq. (5). The proxies separate the methods cleanly: $\hat{L} \approx 47$ for SCRUB, 26 for ORBIT, 16 for Uniform-KL, and 15 for SalUn. Inverting Corollary 1 gives $\epsilon_c \geq (\gamma - L \text{median}(\hat{r}))/2$: the smaller a method’s median radius, the higher the floor on the certificate it can honestly claim, and any tighter certificate is falsified by the audit. The predicted ordering holds—Uniform-KL, with the smallest median radius, is pushed to the highest floor (the weakest honest certificate), while SCRUB’s larger radius leaves its floor low. Because $\hat{L}$ is estimated from the logit-margin Jacobian as a proxy for the probability-margin constant of Corollary 1, I read this as a ranking rather than a measured certificate. SCRUB’s Lipschitz constant is about $3\times$ ORBIT’s and SalUn’s, and this is the same fact that makes its selective gap vanish under Theorem 2: a large $\hat{L}$ lets small perturbations swing the margin both selectively and non-selectively, so the method looks generically fragile rather than selectively leaky.

8.2 Theorem 2: the Pinsker KL table

I compute the forget and retain-control recovery rates ($\hat{p}_F, \hat{p}_R$) for 226 (method, seed, audit-config) rows and report the Pinsker lower bound $KL \geq 2|\hat{p}_F - \hat{p}_R|^2$ (and the tighter Bretagnolle–Huber bound) in `results/analysis/reports/pinsker_kl_table.md`. The ordering is the one the qualitative story predicted, now with a quantitative gap. CE-U recovers $\hat{p}_F = 0.875$ of forget examples against $\hat{p}_R = 0.000$ retain controls ($|\Delta| = 0.875$); SalUn sits near $|\Delta| \approx 0.625$; ORBIT shows a positive but smaller selective gap. SCRUB sits at the bottom with $\hat{p}_F \approx \hat{p}_R \approx 0.97$ and $|\Delta| \approx 0.031$, forcing a KL lower bound of only ≈ 0.002 . By Corollary 2 this formally certifies SCRUB as a *vacuous* selective-leakage case at the audited scale: its strong attack success carries essentially no information about the forget condition, so it is generic adversarial fragility, not unlearning failure.

8.3 Theorem 3: covering implies the mixed-surface failure

The patch, border-frame, and multi-patch results reported above are the empirical witness for Theorem 3. Patch-aware stage-2 closes the single-patch channel on ORBIT and CE-U, but the same defense becomes easier to recover under frame and multi-patch delivery, and mixed-surface stage-2—explicitly trained on multiple surfaces—does not reliably fix the transfer gap. Every coordinate is covered by some 32×32 patch (Lemma 1), so by Theorem 3 worst-case robustness across the family forces constancy at *any* budget—including the $\epsilon \approx 8/255$ the audits use, not only in a large-budget limit (Corollary 3). The mixed-surface failure is therefore not an engineering artifact but the only possible outcome for a non-trivial classifier; the constructive consequence is to retarget local-delivery defense at realistic (non-covering) stochastic delivery priors rather than worst-case invariance.

8.4 Theorem 4: predicted vs. measured certified radius

The smoothed objective of Eq. (10) is implemented as `SmoothedMarginUnlearning` and trained end-to-end on seed 42. Starting from a base ViT-Tiny at 94.2% validation accuracy, a 10-epoch smoothed-margin LoRA fine-tune ($\sigma = 0.10$, $n_{\text{noise}} = 4$, $\gamma = 1.0$) drives forget accuracy $95.7\% \rightarrow 12.0\%$ while retain accuracy is preserved ($93.9\% \rightarrow 93.8\%$). The smoothed certified-radius audit ($n_{\text{noise}} = 256$, $\alpha = 10^{-3}$, 64 forget + 64 retain) certifies a median L_2 radius $R = 0.193$ from Theorem 4. The matched recovery audit gives forget median $\epsilon = 0.00176$ against retain median $\epsilon = 0.00613$ —a $3.5\times$ selective gap. Here the selectivity is carried by the recovery *radius* (the Theorem 1 object) rather than the binary recovery rate of the Pinsker test: both splits saturate near $\hat{p}_F \approx \hat{p}_R \approx 1$ at ϵ_{max} , so the radius separation, not the rate gap, is the informative signal at this budget. Figure 5 joins 128 samples: there are zero retain-side certificate violations, and the ten forget-side “violations” are all clean-leak cases where the un-smoothed model already predicts the forgotten class on the clean input while the smoothed classifier remains robust—a positive signal for smoothing, not a counterexample to the certificate.

9 Defense Discussion

9.1 Feature-subspace stage-2

The first defense family that consistently helps is a post-unlearning feature-subspace stage-2 finetune. Instead of modifying the full unlearning objective from epoch 0, I begin from an already-unlearned adapter and suppress attacked forget-sample features inside a learned recovery subspace. This avoids the instability of earlier logit-margin and teacher-distillation defenses on BalDRO.

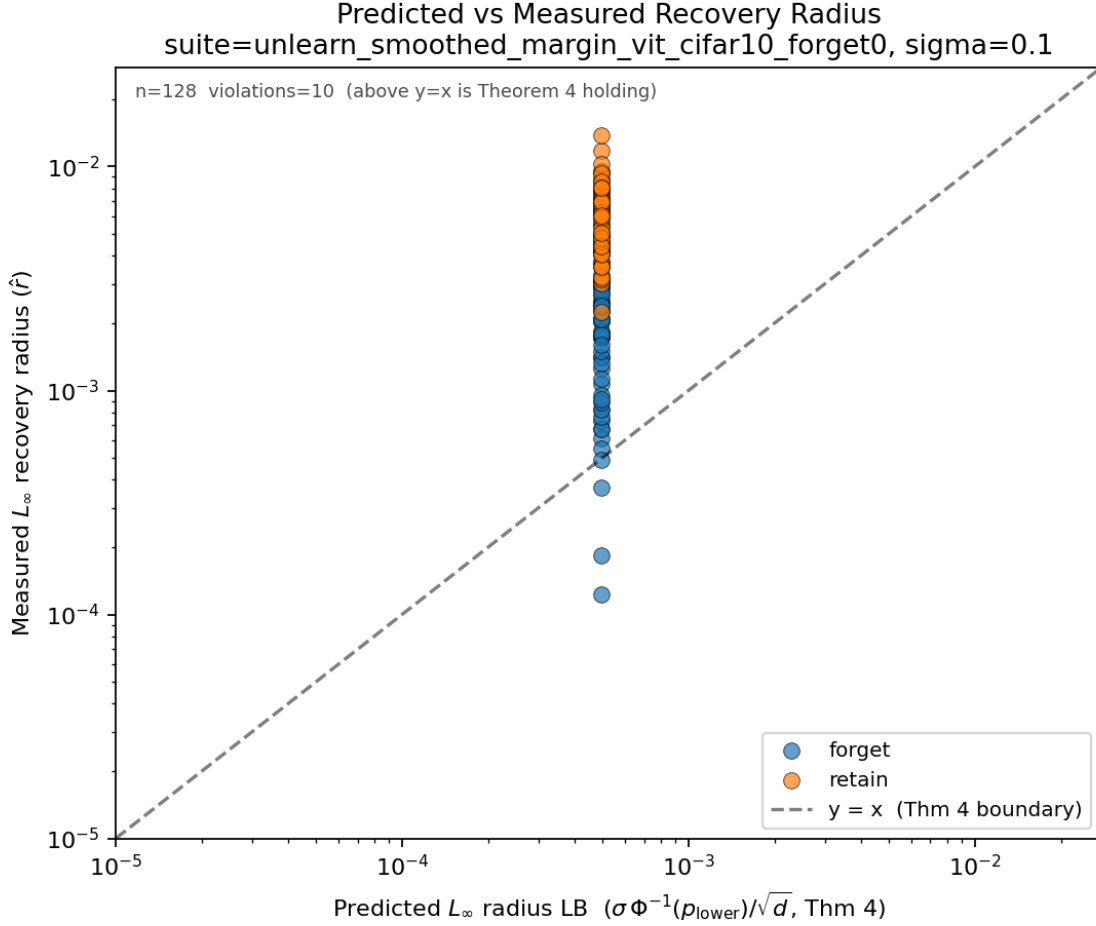


Figure 5: Theorem 4 check. Predicted Cohen–Rosenfeld–Kolter certified radius versus measured recovery radius for the smoothed-margin unlearner (seed 42). Zero retain-side violations; the forget-side points below the line are clean-leak cases on the *un*-smoothed model, which the smoothed classifier corrects.

9.2 Why the gains are small

The absolute changes in reported radii are numerically small because the benchmark itself is small: CIFAR-10 under an L_∞ budget on 224×224 inputs yields radii on the order of a few thousandths. The right question is therefore relative: does the defense move the forget-vs-retain frontier without collapsing clean utility? On BalDRO, SalUn, and the repo-defined ORBIT row, the answer is yes, though modestly.

9.3 What does not generalize

The defense is not universal. RURK remains a hard case under both attack and defense. SCRUB behaves like generic adversarial fragility rather than a clean forgetting-leak regime, so stronger attacks on SCRUB do not necessarily imply a meaningful unlearning failure mode. The newer delivery-surface results make this sharper: patch-aware defenses often fail to transfer to border-frame or multi-patch delivery, and the mixed-surface stage-2 branch does not reliably fix that.

10 Proposed Defense Objective

The experiments suggest that clean-point forgetting is too weak a training target. A more principled direction is targeted-margin adversarial unlearning:

$$\min_{\theta} \mathbb{E}_{x \in R} L_{\text{retain}}(x; \theta) + \lambda_f \mathbb{E}_{x \in F} L_{\text{forget}}(x; \theta) + \lambda_m \mathbb{E}_{x \in F} \max(0, m_t(x + \delta^*(x)) + \gamma), \quad (13)$$

where F is the forget set, R is the retain set, m_t is the forgotten-class target margin, $\delta^*(x)$ is an inner-loop or amortized attack, and γ is a safety margin. This objective is purely empirical: it minimizes a worst-case inner attack and carries no certificate. Theorem 4 upgrades it by replacing the inner attacker $\delta^*(x)$ with Gaussian smoothing of the margin penalty, Eq. (10), which inherits a Cohen–Rosenfeld–Kolter certified recovery radius and (via Corollary 4) a PAC generalization bound to the forget distribution. The end-to-end result in Section 8 (forget 95.7% $\rightarrow$ 12.0%, retain preserved, certified L_2 radius 0.193) is the realized version of this objective. The implemented feature-subspace stage-2 defense can be viewed as a computationally cheaper, uncertified approximation to the same idea: defend against locally recoverable targeted behavior, not just against clean forget accuracy.

11 Discussion

This work began as a multi-regime red-team study, and Section 8 turns that benchmark into the empirical validation of the small theory of Section 3: the Lipschitz–radius scatter tests Theorem 1, the Pinsker KL table tests Theorem 2, the patch/frame/multi-patch transfer failure instantiates Theorem 3, and the smoothed-margin run realizes Theorem 4. Read together, the theory explains *why* the empirical picture is non-universal: recovery radius is lower-bounded by a method-specific Lipschitz constant (so SCRUB’s large $\hat{L}$, which lets perturbations swing the margin freely, is what erases its selective gap), and worst-case multi-surface robustness is provably unattainable for a non-trivial classifier. The evidence does not support a universal “magic” attack or defense. Strong per-example conditional recovery is the main general threat; universal perturbations are mostly weak on the stronger checkpoints; conditional patches expose an additional local attack surface on some methods; border-frame and multi-patch delivery show that local-delivery transfer is a separate problem; robust-base methods define a utility-vs-recovery frontier rather than a binary robust point; and feature-subspace stage-2 provides the first modest defense gains on several targets. Later defense branches remain method-specific: teacher-guided stochastic CVaR helps the repo-defined ORBIT row slightly but not CE-U, while patch-aware stage-2 gives the first clearly positive CE-U result in the patch regime (68.8% $\rightarrow$ 43.8% at 32 $\times$ 32, seed 42) with no retain-side failure, though it does not help on the full-image conditional attack. Mixed-surface stage-2 is informative precisely because it fails: even explicit delivery-surface-aware training does not reliably generalize across one-patch, frame, and multi-patch attacks. Uniform KL reinforces this framing: seed 42 is vulnerable under both the older prompt/probe family and the newer conditional audit, while seed 123 stays flat under the same fast conditional attack. That multifaceted pattern is scientifically stronger than forcing all methods into one attack or defense family.

12 Limitations

I do not claim a universally stronger new optimizer than strong first-order attacks such as `adam_margin` or `apgd_margin`. The amortized short-budget attack is partial and target-dependent. The defense gains are modest and currently demonstrated on a benchmark that is intentionally controlled rather than broad: mostly CIFAR-10 ViT checkpoints, often with LoRA-based unlearning

implementations, plus a smaller number of full-finetune and cross-architecture checks. This makes the benchmark useful for discovering failure modes and transfer gaps, but not yet a direct proxy for most industry deletion settings. Finally, some regimes are better understood as negatives or controls than as headline wins.

The theory carries its own caveats, and I state them rather than hide them. (i) Corollary 1 is rigorous for the bounded probability margin (constant 2, no logit bound), but the Lipschitz constant L it needs is estimated empirically by a logit-margin Jacobian proxy $\hat{L}$ at audit points; a loose proxy still gives a *valid* floor but an indicative rather than measured one, so I read the scatter as a ranking. (ii) Pinsker’s inequality in Theorem 2 is loose; I report the tighter Bretagnolle–Huber bound alongside it, but neither is a hypothesis test with finite-sample coverage at the small per-row audit sizes ($n_F = n_R$ from 4 to 32). (iii) Theorem 3 now holds at *any* budget, including the audit budget, but it constrains only *worst-case* invariance; real defenses target *average-case* robustness over sampled masks, which is exactly why the constructive ρ -thin (non-covering, stochastic) delivery prior, and not worst-case multi-surface training, is the right target. (iv) Theorem 4 certifies an L_2 radius via randomized smoothing, while the recovery audits are L_∞ ; the two are related only up to a $\sqrt{d}$ conversion penalty, and the end-to-end smoothed-margin result is a single-seed (seed 42) demonstration rather than a multi-seed study. Each of these is a concrete next step, not a hidden assumption.

13 Conclusion

Machine unlearning cannot be evaluated reliably with clean forget accuracy alone. This paper makes the per-example recovery radius the central object and shows it supports a small but operational theory: a Lipschitz recovery–certification gap that bounds any claimable certificate (Theorem 1), a Pinsker test that separates selective leakage from generic fragility (Theorem 2), a topological impossibility that explains the mixed-surface transfer failure (Theorem 3), and a smoothed objective with a certified recovery radius (Theorem 4). Across the benchmark, the dominant leakage channel is strong per-example conditional recovery, but local patch, frame, and multi-patch attacks expose additional method-specific surfaces that do not reduce to a single universal vulnerability, and the defense picture is similarly non-universal: feature-subspace stage-2 helps several rows, patch-aware training helps ORBIT and CE-U locally, and worst-case transfer across delivery surfaces is provably unattainable. The benchmark is no longer the headline; it is the experimental validation of the theory.

A Appendix: Seed Variance and Baseline Stability

The clean baseline regime itself is highly variable across seeds for several unlearning methods, especially on forget accuracy. Figure 6 summarizes this instability. This is one reason the paper emphasizes attack-conditioned evaluation rather than clean forget accuracy alone.

B Appendix: Robust-Base Frontier

Figure 2 summarizes the robust-base SalUn operating-point story. At low utility, recovery can be suppressed strongly or disappear on the sampled audit. As retain accuracy increases, conditional recovery re-emerges. This makes robust-base training a frontier choice rather than a single robust point.

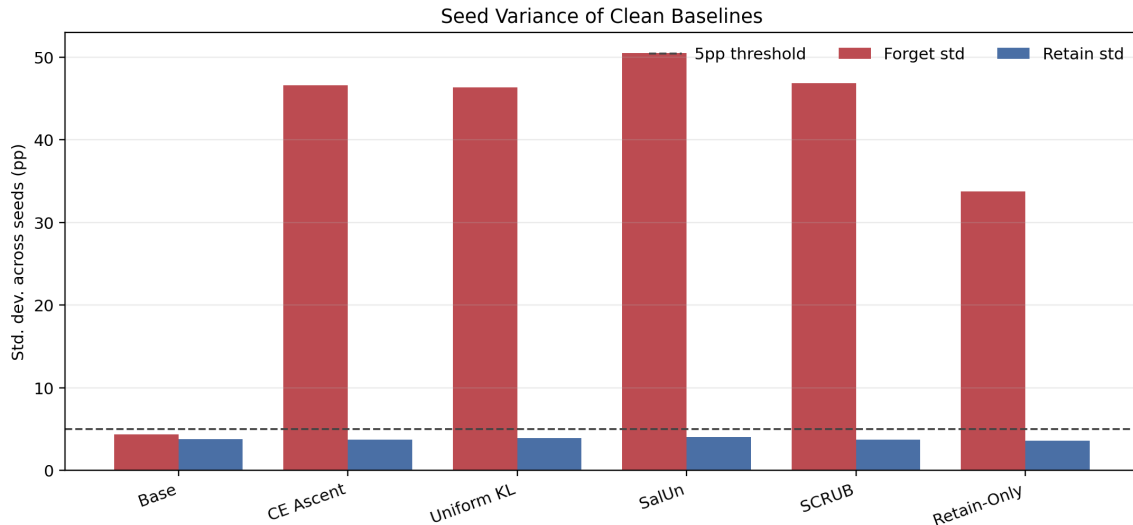


Figure 6: Seed variance of clean baselines. Forget-side variance is large for several methods even when retain-side variance remains small.

C Appendix: Conditional Patch Regime

The patch regime is method-dependent. The repo-defined ORBIT row is the flagship positive case with a 32×32 conditional patch covering 2.04% of the image, strong forget recovery, 0% retain-to-forget success, and limited retain accuracy drop. SalUn is patch-vulnerable only at a larger 48×48 patch budget. BalDRO is effectively negative at the tested budgets. Figure 3 summarizes the patch success rates across methods and seeds.

D Appendix: Delivery-Surface Transfer

The later delivery-surface checks separate one-patch wins from broader local-delivery robustness. Under the conditional frame attack, the ORBIT patch-stage2 row and CE-U combined stage-2 both become easier to recover, and CE-U also incurs retain-side failure. Under conditional multi-patch delivery, the ORBIT patch-stage2 row transfers only weakly, while CE-U combined stage-2 and faithful full-ft SalUn patch-stage2 become easier to recover. Mixed-surface stage-2 was designed to fix this transfer problem, but in practice it only gives narrow gains and often worsens frame or multi-patch robustness. I therefore treat delivery-surface transfer as an open defense problem rather than as solved by the current stage-2 families.

E Appendix: Negative and Hard Cases

RURK remains a hard case for both the amortized attack family and the feature-subspace defense. SCRUB is useful as a negative control: strong conditional attacks succeed, but forget and retain-control radii are almost identical, which suggests generic adversarial fragility rather than a clean forgetting-leak mechanism. Several later defense branches also belong in this bucket: mixed-surface stage-2, frame-transfer failures, and the faithful full-ft SalUn mixed-surface row are all informative mainly because they show what does not transfer.

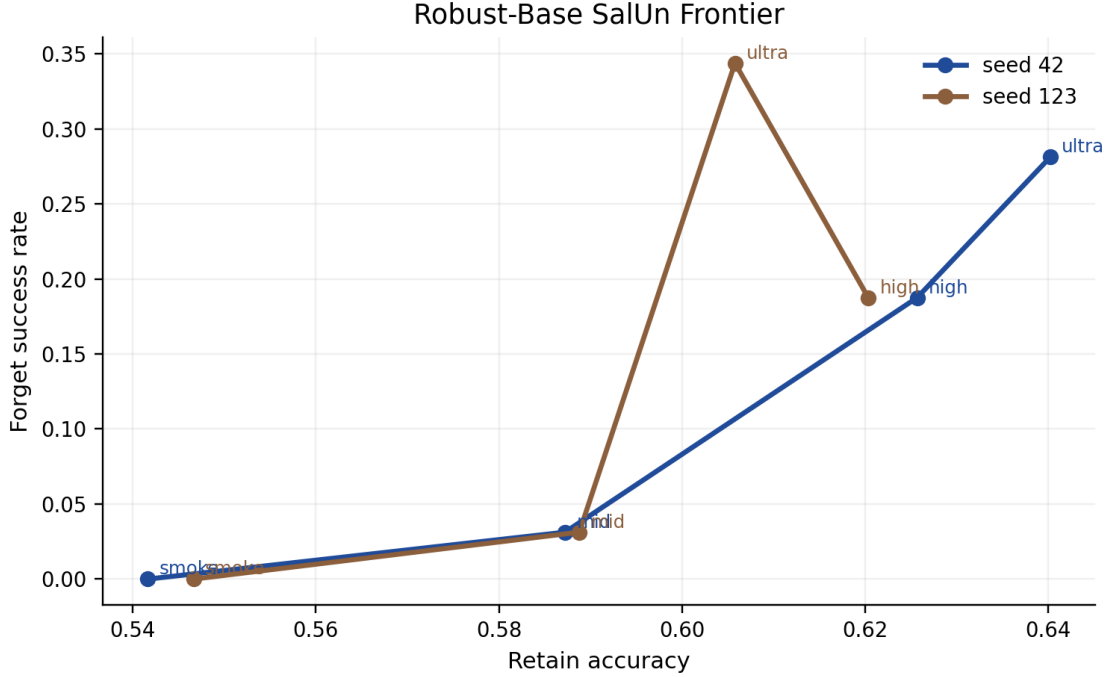


Figure 7: Robust-base SalUn operating-point frontier. Conditional recovery re-emerges as retain accuracy rises, confirming that robust-base training shifts the frontier rather than eliminating leakage.

F Appendix: Reproducibility Artifacts

The paper draws from repository summaries and generated figures in:

- `results/analysis/feature_subspace_tradeoff.md`
- `results/analysis/recovery_tradeoff_summary.md`
- `results/analysis/conditional_patch_summary.md`
- `results/analysis/paper_results_outline.md`
- `results/analysis/figures/`

All figures and tables in this paper are generated from JSON result files committed alongside the code. The repository contains checkpoints for all eight benchmark methods across seeds 42, 123, and 456 together with the audit scripts needed to reproduce each attack regime.

G Appendix: Low-Shot Controls

The lower-shot prompt controls help separate real recovery from trivial prompt memorization. Figure 9 summarizes two useful trends: oracle controls remain flat near zero across the tracked shot counts, while KL and SCRUB can become vulnerable only in specific low-shot or high-shot regimes.

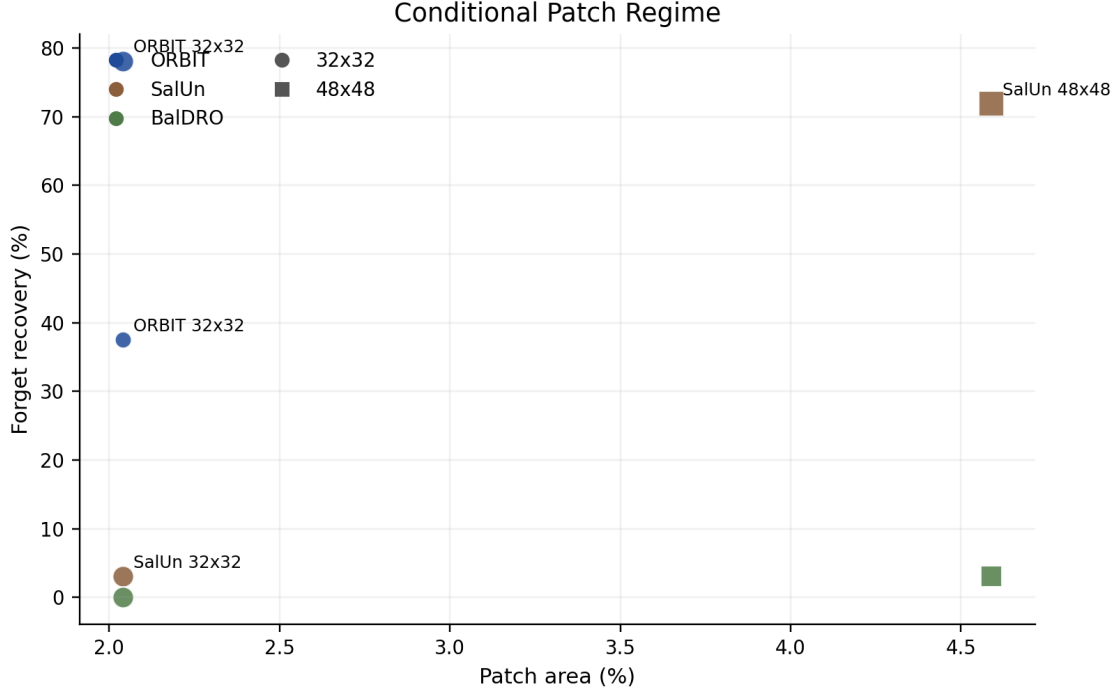


Figure 8: Conditional patch recovery rates across methods. ORBIT is the strongest patch-positive case; BalDRO and robust-base SalUn are effectively negative at 32×32 .

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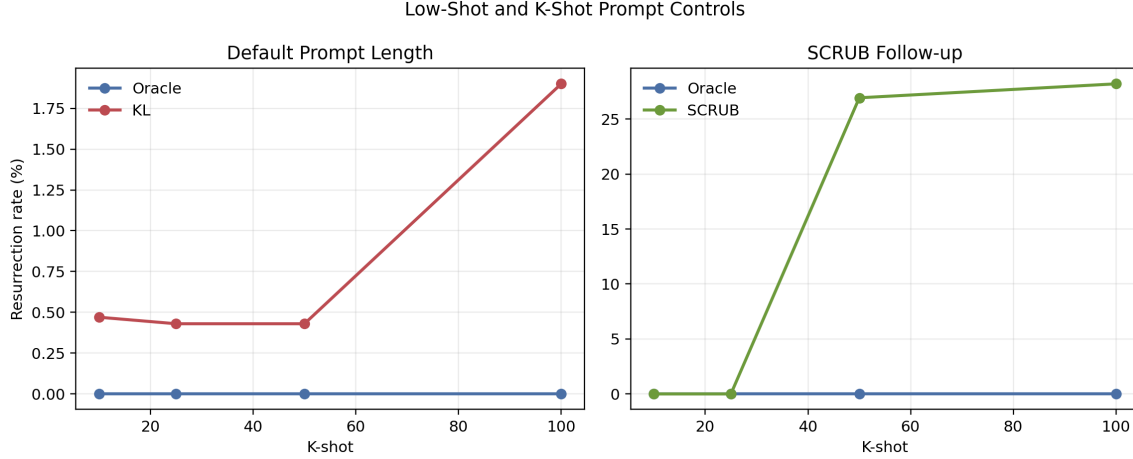


Figure 9: K-shot prompt controls. Oracle stays flat, KL remains weakly vulnerable, and SCRUB only becomes unstable in the larger-shot follow-up.

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